Typed-Budget Object-Relational Abstractions for Imagination-Based Driving Policy Learning

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***Abstract*—Object-relational world models are increasingly used for autonomous driving, but it remains unclear how relation tokens should enter a fixed-capacity latent model and whether representation gains translate to downstream policy learning. In this paper, we study that question through the lens of semantic capacity allocation. We present DOOR, a *d*ecision-*o*riented *o*bject-*r*elational typed-budget abstraction that allocates separate latent budgets to dynamic-agent tokens and relation tokens before feeding them into a shared world model. The key motivation is that naively mixing relations into a shared top-K bottleneck can make relation tokens compete directly with the dynamic agents that driving decisions must track. Under a fair 16-slot budget on nuScenes, the proposed abstraction improves representation sufficiency substantially over an object-only baseline, reducing rare-agent displacement error by 55.2% and improving interaction recall at 1 m by 8.3 points, while naive relation mixing collapses. We then study whether this representation benefit transfers to downstream policy optimization. The answer is not uniform across datasets. On nuScenes short-horizon latent imagination, the object-only policy remains the most stable baseline. On planning-oriented nuPlan, however, the no-visibility decoupled variant is the robust winner at both 20k and 50k scale. At 50k, it achieves 14.51 ± 2.93 return and 0.259 ± 0.045 collision rate, versus 1.72 ± 17.89 and 0.610 ± 0.146 for object-only. An additional 50k offline planner-like sanity probe also shows lower teacher-action error (6.63 ± 0.11 versus 8.86 ± 0.37), indicating that the gain is not limited to latent reward. Follow-up cross-dataset transfer rollouts and nuScenes rollout-horizon sensitivity further support the same interpretation. These results show that typed-budget relation-aware abstractions are reliably useful at the representation level, but their downstream benefit depends on the planning regime and tokenization characteristics.**

***Index Terms*—Autonomous driving, world models, object-relational representation, semantic abstraction, policy learning.**

I. Introduction

WORLD models offer an appealing route to autonomous driving policy learning because they replace expensive environment interaction with imagined trajectories [1], [2]. For driving, this promise is especially attractive: direct policy optimization is expensive, safety-critical, and often bottlenecked by sparse feedback. At the same time, a driving policy must reason about a small number of highly consequential entities and their relations rather than an undifferentiated global state.

This tension raises a representation question that is still under-specified in the current literature: *how should relation-aware structure enter a fixed-capacity latent driving model?* A common answer is to append relation tokens to the same top-K bottleneck used for dynamic agents. However, that design implicitly assumes all tokens should compete for the same limited latent budget. In autonomous driving, this assumption is brittle. A relation token that encodes lane conflict or time-to-collision may be useful only if the dynamic agents involved in that interaction are still represented in the latent state.

We study this problem through the lens of semantic capacity allocation. Our central claim is that relation-aware driving world models fail or succeed not only because of what information they encode, but also because of *how the latent budget is divided across semantic roles*. This motivates DOOR, a decision-oriented object-relational typed-budget abstraction that allocates separate latent budgets to dynamic-agent tokens and relation tokens before feeding them into a shared world model. The design is simple, but the hypothesis is specific: if relation tokens are mixed naively into a shared bottleneck, they will compete with dynamic agents for scarce slots and degrade decision-sufficient structure.

The second question is whether representation gains translate to policy learning. Better forecasting metrics do not automatically imply a better control landscape for latent-space policy optimization. We therefore organize the paper in two stages. Stage 0 studies representation sufficiency under a fair 16-slot budget on nuScenes. Stage 1 studies imagination-based policy learning across two downstream regimes: nuScenes short-horizon latent imagination and a planning-oriented nuPlan preprocessed benchmark. This split allows us to ask not only whether typed-budget abstraction is useful, but also when it becomes useful for policy optimization.

The resulting picture is more nuanced and, we believe, more valuable than a simple "method always wins" story. On nuScenes, Stage 0 strongly favors the decoupled typed-budget representation, but Stage 1 still prefers the object-only policy baseline. On nuPlan 20k and its 50k scale-up, however, the ranking flips: the decoupled no-visibility variant becomes the most reliable policy-learning condition, and an additional offline planner-like sanity probe supports the same choice. Cross-dataset transfer rollouts and horizon-sensitivity diagnostics point to the same conclusion. This shows that relation-aware abstraction is not a universal free lunch. Its benefit depends on whether the downstream planning regime and tokenization characteristics make relation structure actionable for policy learning.

Under this scope, the paper is not a full end-to-end perception-to-control system and not yet a full external closed-loop benchmark paper. It is a method-and-analysis paper about decision-sufficient semantic abstraction for imagination-based driving policy learning. That narrower framing is deliberate:

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it lets us isolate the representation-policy interface before introducing additional confounders such as perception noise or simulator-specific artifacts.

Figure 1 summarizes this overall story: a fixed-budget failure mode in Stage 0, a structural fix through typed dyn/rel budgeting, and a downstream ranking reversal that only appears once the planning regime makes relation structure actionable.

The contributions of this paper are as follows.

1) We identify *inter-type budget competition* as a failure mode of relation-aware latent driving models: naively mixing relation tokens into a shared bottleneck can degrade dynamic-agent coverage under a fixed latent budget.
2) We propose DOOR, a typed-budget object-relational abstraction that decouples dynamic-agent and relation-token selection while keeping the total world-model context fixed.
3) We show that typed-budget abstraction consistently improves representation sufficiency on nuScenes, but its downstream policy-learning benefit is dataset-dependent: object-only remains the most stable baseline on nuScenes Stage 1, whereas the no-visibility decoupled variant becomes the robust winner on nuPlan 20k/50k, in downstream offline planner-like probes, and in follow-up transfer diagnostics.

II. Related Work

A. World Models for Driving Policy Learning

World models have become a central approach to imagination-based policy learning [1], [2]. In driving, they promise to replace expensive simulator interaction with learned latent rollouts. Yet most formulations focus on latent prediction efficiency rather than on how the latent budget should be allocated across semantic roles. Our work addresses that missing question directly.

B. Object-Centric and Relational Representations

Object-centric learning [8] has shown that scenes composed of interacting entities benefit from structured representations. Autonomous driving inherits this need: policies must reason about nearby vehicles, vulnerable road users, and relation cues such as right-of-way or lane conflict. Prior work has established that object and map structure help prediction and planning. We extend that line by asking how object and relation structure should enter a fixed-capacity world model for downstream policy learning.

C. Driving Benchmarks and Cross-Dataset Evaluation

nuScenes [4] and nuPlan [5] provide two complementary views of driving data: the former is widely used for perception and trajectory modeling, whereas the latter is more directly aligned with planning evaluation. Additional platforms such as CARLA [3], World on Rails [9], Waymax [6], and NAVSIM [7] further emphasize that downstream task setting matters. Our experiments leverage this contrast directly: the same representation can look very different once it is used for imagination-based policy learning under different planning regimes.

III. Problem Formulation

We consider a driving agent acting in a partially observable traffic environment. At time step t, the agent receives a structured scene representation

$$\mathbf{x}_t = \{\mathbf{e}_t, \mathcal{O}_t, \mathcal{M}_t, \mathcal{R}_t\}, \tag{1}$$

where $\mathbf{e}_t$ denotes the ego token, $\mathcal{O}_t$ denotes dynamic object tokens, $\mathcal{M}_t$ denotes map tokens, and $\mathcal{R}_t$ denotes relation tokens. The policy $\pi(\mathbf{a}_t \mid \mathbf{x}_t)$ outputs a driving action $\mathbf{a}_t$, which may correspond to low-level control or a compact motion command.

The world model cannot consume the full scene indefinitely. It must compress $\mathbf{x}_t$ into a latent context of fixed capacity K. The central problem studied in this paper is therefore not only *what* information to encode, but also *how* to allocate capacity between dynamic-agent information and relation information. If relation tokens enter the same bottleneck as dynamic-agent tokens, they may displace the very entities needed for downstream control. Our objective is to learn an abstraction A_θ and world model f_θ such that the resulting latent state remains decision-sufficient for two downstream tasks:

1) **Representation sufficiency:** preserve dynamic prediction, rare-agent tracking, and interaction-critical structure under a fair fixed latent budget.
2) **Policy learning:** support stable imagination-based policy optimization.

A. Scope and Evaluation Assumptions

This work studies decision learning under structured scene tokens rather than full end-to-end perception. We assume *oracle-token evaluation*: scene tokens are derived from annotations, simulator state, or an equivalent trusted front-end. This assumption is intentional. It isolates the abstraction-and-policy question from perception noise and allows us to compare downstream planning regimes directly. The resulting paper should therefore be read as a study of latent representation and imagination-based policy learning, not as a claim about complete perception-to-control autonomy.

IV. Method

A. Scene Tokenization

Each scene is represented in an ego-centric local frame to encourage cross-benchmark consistency. The raw feature layout used by the current implementation contains a fixed-length token vector of dimension $D_{\text{raw}} = 40$, with the first coordinates reserved for relative position, velocity, object extent, risk proxies, visibility, time-to-collision, lane conflict, interaction priority, distance, heading, and token type. The maximum token budget is $S = 97$, corresponding to one ego token, a bounded set of dynamic-object tokens, map tokens, relation tokens, and padding.

This design serves two purposes. First, it keeps the model interface stable across offline datasets, reactive simulators, and high-fidelity evaluators. Second, it exposes relation features explicitly instead of forcing the world model to infer all interaction structure from a holistic scene embedding. In practice,

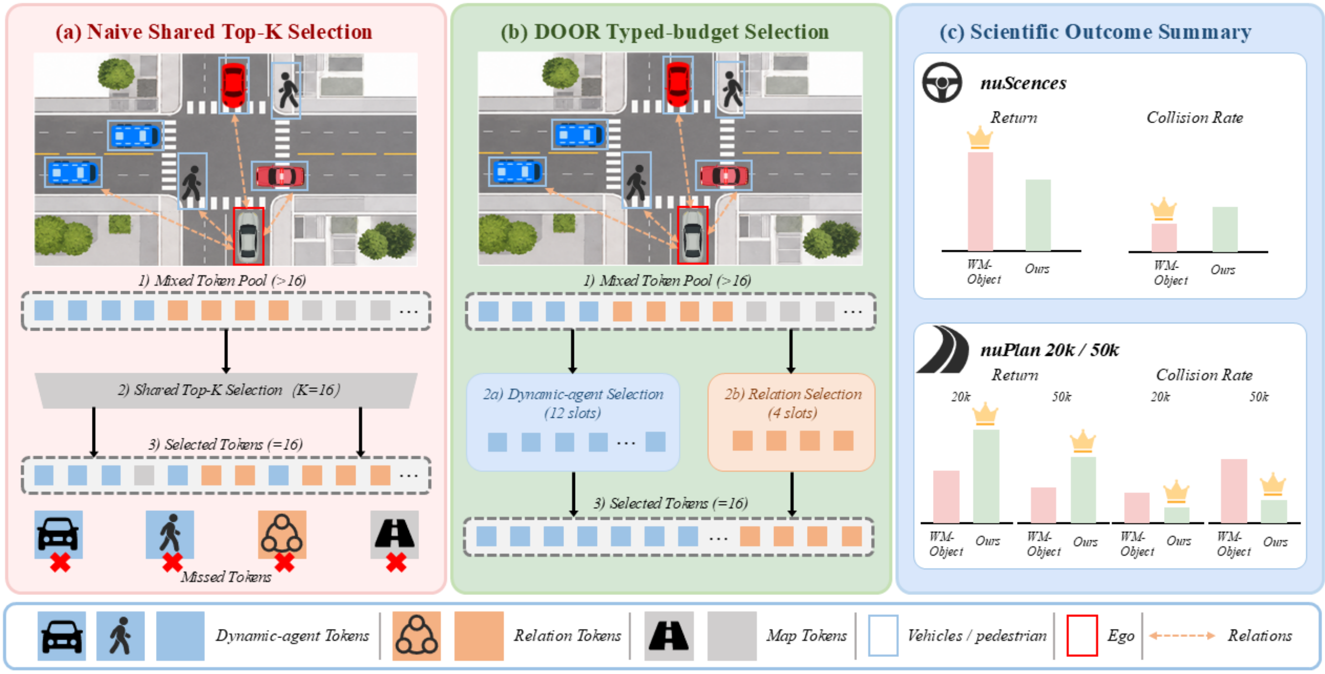

Fig. 1. Teaser figure summarizing the paper's core story. Under a fixed 16-slot latent budget, naive shared object-relation selection can displace decision-critical dynamic agents; typed-budget decoupling restores that context; and the resulting policy-learning benefit appears only in the downstream regimes where relation structure becomes actionable.

relation tokens encode pairwise or locally aggregated cues such as relative position, relative velocity, distance, time-to-collision proxy, lane conflict indicator, visibility or occlusion score, and interaction priority.

The token schema is shared across the two main Stage-1 regimes, but the token statistics are not. nuScenes tokens are generated from annotated driving scenes with visibility variation, whereas the nuPlan preprocessed pipeline materializes one NPZ anchor frame per sample and the official nuPlan closed-loop adapter recomputes relation tokens online from current ego and tracked-object geometry. In that adapter, relation tokens are constructed from relative geometry, relative velocity, TTC, lane conflict, risk, and priority heuristics, while visibility is effectively constant. This difference is useful later when interpreting why visibility weighting behaves differently on nuScenes and nuPlan.

B. Decision-Sufficient Abstraction

Given token features $\mathbf{x}_{t,i}$ and token types $c_{t,i}$, an encoder produces latent token embeddings

$$\mathbf{h}_{t,i} = g_{\text{enc}}(\mathbf{x}_{t,i}, c_{t,i}), \tag{2}$$

where g_{enc} is implemented as a token encoder with learned type embeddings. A decision-sufficient abstraction module then scores tokens using ego-conditioned attention:

$$\alpha_{t,i} = \text{softmax}_i \left(\frac{(\mathbf{W}_q \mathbf{h}_{t,\text{ego}})^\top (\mathbf{W}_k \mathbf{h}_{t,i})}{\sqrt{d}} \right), \tag{3}$$

followed by a learned importance head

$$s_{t,i} = g_{\text{score}}([\mathbf{h}_{t,i}; \alpha_{t,i}]). \tag{4}$$

The top-K tokens, together with rule-based safety retention for critical cases such as low time-to-collision or high-conflict pedestrians, form the selected latent entity set $\mathbf{z}_t^{\text{sel}}$. In the current implementation, $K = 16$. A pooled global latent $\mathbf{z}_t^{\text{glob}}$ is computed in parallel for policy evaluation and value prediction.

This abstraction is not merely a compression stage. It acts as a decision bottleneck: if the representation fails to retain safety-critical interactive structure, the downstream policy should degrade in precisely the scenarios that matter to driving. Our experimental design therefore treats representation sufficiency as a central scientific question rather than a minor implementation detail.

In the final DOOR instantiation used in our Stage-0 representation study, we further decouple this bottleneck by semantic role. Instead of forcing dynamic-agent tokens and relation tokens to compete inside a single shared top-K pool, we allocate typed budgets K_{dyn} and K_{rel} such that

$$K_{\text{dyn}} + K_{\text{rel}} = K. \tag{5}$$

The world model then receives the concatenation of the selected dynamic and relation slots. Under the fair 16-slot budget used in our experiments, we set $(K_{\text{dyn}}, K_{\text{rel}}) = (12, 4)$. This design is motivated by a simple failure mode: when relation tokens are mixed naively into a shared bottleneck, they may displace the dynamic-agent slots required for forecasting and interactive decision making. The typed-budget formulation preserves the total context size while avoiding inter-type budget competition.

Figure 2 is intended to make this distinction visually explicit by contrasting shared top-K selection with typed-budget decoupling inside the same overall world-model pipeline.

C. Reactive Object-Relational World Model

The world model receives the selected token set $\mathbf{z}_t^{\mathrm{sel}}$ and action $\mathbf{a}_t$, then predicts next-step latent entities, reward, continuation probability, and collision risk:

$$\hat{\mathbf{z}}_{t+1}^{\mathrm{sel}}, \hat{r}_t, \hat{c}_t, \hat{\rho}_t = f_\theta(\mathbf{z}_t^{\mathrm{sel}}, \mathbf{a}_t). \tag{6}$$

The action is injected into token dynamics so that predicted transitions can depend on ego intervention. This is essential: without action-conditioned multi-agent prediction, the latent model would regress toward a sophisticated replay predictor rather than a reactive closed-loop model.

The training loss combines next-token prediction, reward prediction, continuation prediction, and auxiliary collision-risk prediction:

$$\mathcal{L}_{\mathrm{wm}} = \lambda_{\mathrm{obs}}\mathcal{L}_{\mathrm{obs}} + \lambda_r\mathcal{L}_r + \lambda_c\mathcal{L}_c + \lambda_{\mathrm{col}}\mathcal{L}_{\mathrm{col}}. \tag{7}$$

The current scaffold also allows an optional behavior-cloning regularizer during warm start,

$$\mathcal{L} = \mathcal{L}_{\mathrm{wm}} + \lambda_{\mathrm{bc}}\mathcal{L}_{\mathrm{bc}} + \mathcal{L}_{\mathrm{pol}}, \tag{8}$$

which stabilizes early training when the world model and value function are still inaccurate.

D. Latent Imagination Policy Learning

We optimize a policy-and-value stack on imagined trajectories produced by the world model. At each rollout step, the policy head reads $\mathbf{z}_t^{\mathrm{glob}}$ and outputs the parameters of a continuous diagonal-Gaussian action distribution. To prevent imagination training from exploding numerically, the action mean is bounded with a scaled hyperbolic tangent and the log standard deviation is clamped to a finite interval before sampling. A value head estimates state value from the same global latent. In Stage 1, we use a 5-step latent rollout, GAE-λ targets, reward clipping, detached world-model updates during policy optimization, and a real-batch sanity loss that keeps the representation grounded. The purpose of this stage is not to claim a new optimization algorithm, but to test whether a representation that looks stronger in Stage 0 also produces a more favorable optimization landscape for imagination-based policy learning.

The resulting training procedure is summarized in Algorithm 1. Compared with render-in-the-loop optimization, this design keeps policy improvement in latent space and lets us study directly whether semantic abstraction quality changes the stability of downstream policy optimization.

Algorithm 1 DOOR Policy-Learning Procedure

1: Collect or load structured scene transitions from offline data and reactive rollouts.
2: Encode tokens and compute decision-sufficient latent entities.
3: Update the world model using $\mathcal{L}_{\mathrm{wm}}$ in (7).
4: Warm start the policy with behavior cloning if expert actions are available.
5: Roll out imagined latent trajectories with the current policy and world model.
6: Update the policy and value heads using imagined rewards, continuation, and value targets.
7: Periodically evaluate the policy in external closed-loop benchmarks.
8: Keep the final model based on external evaluation rather than latent-only validation.

TABLE I
EXPERIMENTAL PROTOCOLS USED IN THE PAPER

Protocol	Dataset and Split	Purpose
Stage 0	nuScenes, 700 scenes, 28 096 samples	Representation sufficiency
Stage 1A	nuScenes, 700 scenes, 5 622 val samples	Short-horizon imagination training
Stage 1B	nuPlan 20k, 16k/4k split	Planning-oriented imagination training
Stage 1C	nuPlan 50k, 40k/10k split	Main scale-up and downstream probe

V. EXPERIMENTAL SETUP

A. Datasets and Protocols

Table I summarizes the protocols used in the paper. Stage 0 uses nuScenes for representation analysis under a fair 16-slot world-model budget. Stage 1 reuses the same abstraction family for imagination-based policy learning on nuScenes and on planning-oriented nuPlan preprocessed benchmarks at 20k and 50k scale. We additionally evaluate the strongest nuPlan condition with an offline planner-like sanity probe on the 50k validation split and an interaction-conditioned subset analysis over lane-conflict, low-TTC, and dense-agent subsets. As support diagnostics rather than new benchmark claims, we also run pure cross-dataset rollout evaluation and a nuScenes rollout-horizon sensitivity study. The nuPlan Stage-1 experiments use Stage-0 warm starts trained on the corresponding dataset split.

B. Variants

Stage 0 compares six compact 16-slot variants together with one 97-token upper-bound reference:

1) **Holistic-16Slot:** learned-query compression from the full token set.
2) **Object-Only-16:** top-K selection over ego and dynamic-agent tokens only.
3) **Object+Relation-16:** naive shared-budget top-K over dynamic and relation tokens.
4) **Object+Relation+Visibility-16:** shared-budget relation-aware selection with visibility weighting.
5) **Obj+Rel-Decoupled:** typed-budget abstraction with 12 dynamic slots and 4 relation slots.
6) **Decoupled+Visibility:** typed-budget abstraction plus visibility weighting on the dynamic path.

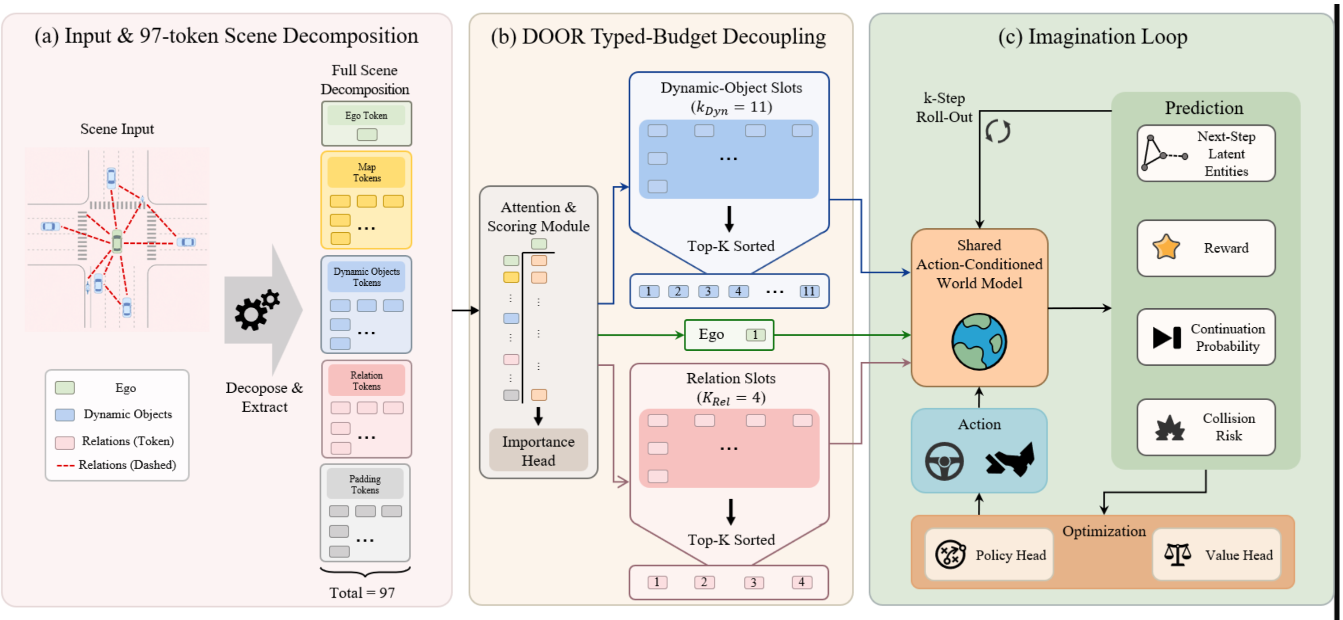

Fig. 2. Method overview of DOOR. The figure contrasts naive shared-budget selection with the proposed typed-budget dyn/rel selection path, annotates the fair $(12, 4)$ split inside the shared 16-slot budget, and shows how the selected slots feed the action-conditioned latent dynamics model and imagination-based policy/value optimization loop.

Stage 1 focuses on the conditions most relevant to policy learning: `wm_object`, `wm_decoupled`, and `wm_decoupled_no_vis`. Additional nuScenes-only ablations (14+2 typed budget and relation-to-critic-only fusion) are discussed in text as diagnostic support.

C. Metrics

Stage 0 reports four representation-centric metrics.

1) **Dyn Rollout MSE:** nearest-match squared error on (x, y, v_x, v_y) for dynamic agents.
2) **Collision F1:** binary F1 of predicted collision risk against labels derived from relation-token time-to-collision below 3 s.
3) **Rare ADE:** nearest-match displacement error on pedestrians and cyclists.
4) **Interaction Recall @ 1 m:** fraction of rare agents within 20 m of ego whose nearest-match error is below 1 m.

Stage 1 reports imagination-policy metrics:

1) **Return:** K-step imagined latent reward sum.
2) **Collision Rate:** fraction of imagined rollouts whose collision probability exceeds threshold.
3) **Collision Mean:** mean collision probability over the rollout.
4) **Ego Stability (cos):** cosine distance between successive ego latents, used as a liveness and divergence sentinel rather than a standalone quality metric.

For the downstream nuPlan 50k sanity probe, we additionally report teacher-action mean-squared error and action ΔL_2 distance to characterize how closely the learned policy aligns with an offline planner-like behavior target. The same Stage-1 metrics are reused for the cross-dataset and horizon-sensitivity follow-ups.

D. Implementation Details

The shared model configuration uses raw token dimension 40, model dimension 128, hidden dimension 256, four attention heads, two transformer world-model layers, and a token budget of 97. The fair compact setting uses $K = 16$ latent slots. Stage 0 uses Adam with learning rate 10^{-3}, batch size 32, 15 epochs, and three seeds $\{7, 42, 2026\}$. Stage 1 uses a 5-step imagination horizon, batch size 128, learning rate 4×10^{-3}, 10 epochs, three seeds $\{7, 42, 123\}$, GAE-λ targets with $(\gamma, \lambda) = (0.97, 0.95)$, reward clipping to $[-5, 5]$, a Huber critic, detached world-model updates, and bounded action statistics with clipped sampled actions in $[-5, 5]$.

VI. Results

A. Stage-0 Representation Sufficiency on nuScenes

The representation ablation in Table II is the scientific center of the paper. Unlike a standard ablation that only asks whether more structure helps, this experiment asks *how* relation structure should enter a fixed-capacity world model. The key question is whether relation tokens improve decision-sufficient forecasting, or whether they simply consume the slots needed to represent dynamic agents. To keep that question visually clear, the main table reports only the *fair 16-slot comparison*. The 97-token holistic model is treated separately as a non-fair upper bound.

The headline result is positive: under the same 16-slot budget, the proposed typed-budget models are the strongest compact variants on the metrics most directly tied to interaction-critical representation. Relative to Object-Only-16, Obj+Rel-Decoupled reduces Rare ADE by 55.2% and improves Interaction Recall@1m by 8.3 points, while Decoupled+Visibility further lowers Dyn Rollout error by 49.9%. Compared with Holistic-16Slot, the decoupled model improves Interaction

TABLE II

STAGE-0 REPRESENTATION SUFFICIENCY ABLATION ON NUSCENES UNDER A FAIR 16-SLOT WORLD-MODEL CONTEXT BUDGET. ALL RESULTS ARE AVERAGED OVER THREE SEEDS. BOLDFACE MARKS THE BEST FAIR-BUDGET RESULT IN EACH COLUMN.

Representation	Ctx	Dyn Rollout ↓	Collision F1 ↑	Rare ADE ↓	IntRec@1m ↑
Holistic-16Slot	16	2.11 ± 0.16	0.978 ± 0.011	1.42 ± 0.01	0.643 ± 0.015
Object-Only-16	16	3.74 ± 1.01	0.946 ± 0.004	1.10 ± 0.12	0.901 ± 0.034
Object+Relation-16 (naive)	16	40.28 ± 29.54	0.980 ± 0.013	7.51 ± 5.48	0.430 ± 0.407
Object+Relation+Visibility-16	16	15.80 ± 9.93	0.933 ± 0.064	2.96 ± 1.64	0.728 ± 0.155
Obj+Rel-Decoupled (Ours)	16	$\mathbf{2.11 \pm 0.19}$	0.929 ± 0.039	$\mathbf{0.49 \pm 0.18}$	$\mathbf{0.984 \pm 0.014}$
Decoupled+Visibility (Ours)	16	$\mathbf{1.88 \pm 0.23}$	0.926 ± 0.029	0.52 ± 0.05	0.979 ± 0.008
Δ Obj+Rel-Decoupled vs Object-Only	–	-43.5%	-1.8 pts	-55.2%	$+8.3$ pts
Δ Decoupled+Vis vs Object-Only	–	-49.9%	-2.1 pts	-52.6%	$+7.8$ pts

Recall@1m from 0.643 to 0.984, a gain of 34.1 points. These are the gains that matter most for the paper's claim: relations become useful once the abstraction budget matches semantic role.

The second result explains *why* this advantage appears. Naively mixing relation tokens into a shared 16-slot bottleneck fails catastrophically. Compared with Object-Only-16, the naive Object+Relation-16 variant increases Dyn Rollout error from 3.74 to 40.28, increases Rare ADE from 1.10 to 7.51, and drops Interaction Recall@1m from 0.901 to 0.430. This behavior is also highly unstable across seeds, with an Interaction Recall standard deviation of 0.407. The implication is not that relation structure is harmful, but that a shared bottleneck creates inter-type budget competition that starves the model of dynamic-agent slots. Fig. 3 visualizes this mechanism directly: the naive shared-budget variants allocate too many slots to relation tokens, whereas the decoupled variants preserve the intended dynamic-versus-relation budget split.

The third result is a transparent trade-off. The decoupled variants give up a small amount of collision-prediction F1 compared with Object-Only-16 (0.929 versus 0.946), which we attribute to the strict relation-slot budget of four under the fair protocol. We keep this trade-off visible because it makes the design interpretable: the typed-budget abstraction primarily improves dynamic-agent forecasting and interaction recall, while collision reasoning remains competitive but not dominant at this stage.

For completeness, a non-fair 97-token holistic upper bound achieves Dyn Rollout 0.11 ± 0.12, Collision F1 0.988 ± 0.006, Rare ADE 0.26 ± 0.02, and IntRec@1m 1.000 ± 0.000. We do not place this reference in the main table because it uses more than six times the latent context of the compact variants. Its role is only to indicate the remaining headroom above the fair 16-slot setting.

B. Stage-1 Cross-Dataset Imagination Policy Learning

Stage 0 establishes that typed-budget abstraction preserves decision-critical dynamic information under a fixed latent budget, but it does not yet show whether this representation supports policy improvement. Table III therefore studies imagination-based policy optimization across two downstream regimes: nuScenes short-horizon latent imagination and planning-oriented nuPlan imagination at 20k and 50k scale. This table is the core Stage-1 result of the paper.

The first message of Table III is cautionary. Better representation does not automatically imply better policy learning. On nuScenes, the object-only policy remains the strongest and most stable baseline among the three main Stage-1 conditions. This overturns the simplistic interpretation that typed-budget abstraction should dominate everywhere merely because it dominates Stage 0.

The second message is that this is not a global failure of decoupled abstraction. On nuPlan 20k, the ranking flips decisively, and the same pattern survives the 50k scale-up. At 20k, WM-Decoupled-NoVis achieves 17.50 ± 1.37 return versus 4.74 ± 13.95 for WM-Object, while also lowering collision rate from 0.373 ± 0.083 to 0.226 ± 0.105. At 50k, the same condition remains positive on all three seeds and retains a large safety gap, achieving 14.51 ± 2.93 return and 0.259 ± 0.045 collision rate versus 1.72 ± 17.89 and 0.610 ± 0.146 for WM-Object. This makes the nuPlan result much stronger than an isolated best-seed comparison.

The third message concerns visibility weighting. In Stage 0, visibility weighting slightly improves the best dynamic-rollout metric. In Stage 1, however, its role is dataset-dependent. On nuScenes, the no-visibility condition is the weakest of the three main policy-learning variants. On nuPlan 20k and 50k, the no-visibility decoupled condition is the strongest. This suggests that visibility should not be treated as a universally helpful enhancement; its effect depends on how the downstream tokenization and planning regime use that information.

Two auxiliary nuScenes ablations support this interpretation. Shrinking the typed budget from $(12, 4)$ to $(14, 2)$ does not rescue Stage-1 performance (2.47 ± 4.14 return, 0.808 ± 0.196 collision rate), and routing relation information only to the critic yields only partial improvement (5.72 ± 7.25 return, 0.729 ± 0.169 collision rate). Together, these results suggest that the nuScenes Stage-1 gap is not explained by a trivial over-allocation of relation slots or by a simple actor-versus-critic fusion mistake.

As a supporting warm-start result, Stage 0 on nuPlan 20k and 50k also strongly favors decoupled abstraction. At 20k, object-only yields Dyn Rollout 168.733 and Rare ADE 17.981,

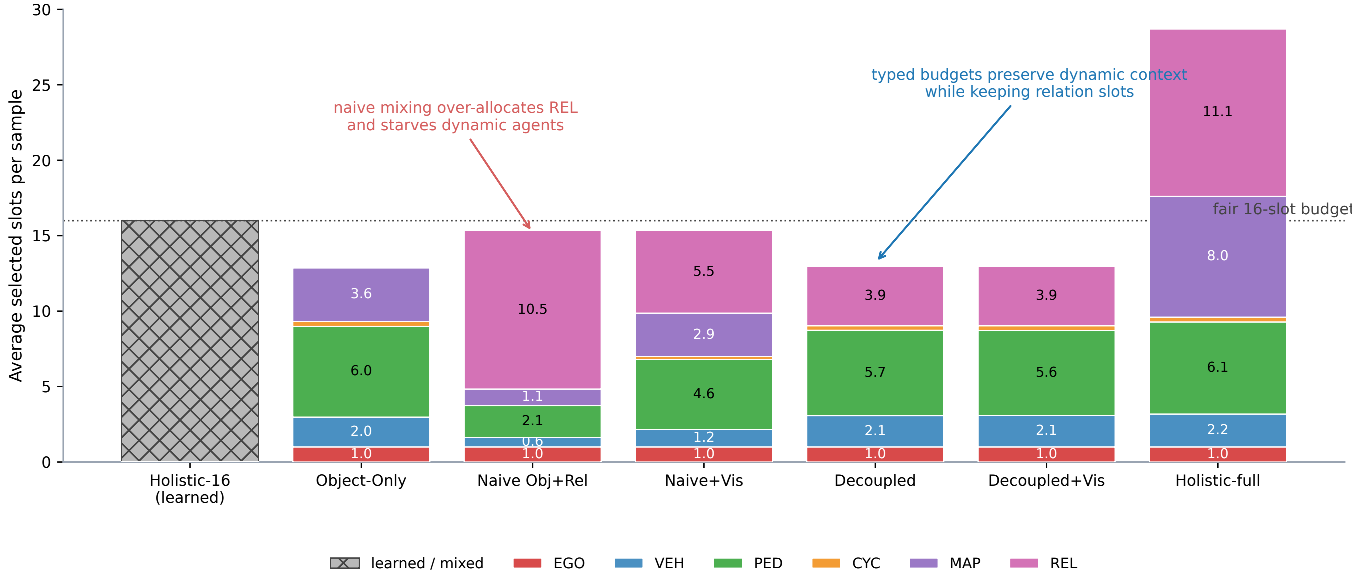

Fig. 3. Stage-0 slot-type composition over validation samples, drawn directly from aggregate evaluation data. Under the fair 16-slot protocol, naive shared object-relation selection allocates too much of the compact context to relation tokens and starves nearby dynamic agents. The proposed typed-budget variants preserve dynamic coverage while still reserving a dedicated relation budget.

TABLE III
STAGE-1 IMAGINATION-BASED POLICY LEARNING ACROSS DATASETS AND NUPLAN SCALES. ALL RESULTS ARE AVERAGED OVER THREE SEEDS. BOLDFACE MARKS THE BEST CONDITION WITHIN EACH DATASET BLOCK.

Dataset	Condition	Return ↑	Coll. Rate ↓	Coll. Mean ↓	Ego Stability (cos)
nuScenes	**WM-Object**	$\mathbf{31.79 \pm 19.70}$	$\mathbf{0.597 \pm 0.048}$	$\mathbf{0.610 \pm 0.048}$	0.258 ± 0.058
nuScenes	WM-Decoupled	4.34 ± 13.66	0.695 ± 0.283	0.676 ± 0.243	0.636 ± 0.108
nuScenes	WM-Decoupled-NoVis	0.34 ± 15.97	0.820 ± 0.260	0.814 ± 0.167	$\mathbf{0.223 \pm 0.033}$
nuPlan 20k	WM-Object	4.74 ± 13.95	0.373 ± 0.083	0.395 ± 0.057	0.426 ± 0.099
nuPlan 20k	WM-Decoupled	13.48 ± 4.09	0.488 ± 0.217	0.509 ± 0.193	0.546 ± 0.046
nuPlan 20k	**WM-Decoupled-NoVis**	$\mathbf{17.50 \pm 1.37}$	$\mathbf{0.226 \pm 0.105}$	$\mathbf{0.251 \pm 0.101}$	$\mathbf{0.136 \pm 0.022}$
nuPlan 50k	WM-Object	1.72 ± 17.89	0.610 ± 0.146	0.614 ± 0.113	0.367
nuPlan 50k	**WM-Decoupled-NoVis**	$\mathbf{14.51 \pm 2.93}$	$\mathbf{0.259 \pm 0.045}$	$\mathbf{0.277 \pm 0.033}$	**0.222**

whereas both decoupled variants reduce these to 3.210 and 0.848. At 50k, object-only yields Dyn Rollout 116.627, Rare ADE 17.318, and IntRec@1m 0.006, whereas the decoupled model achieves 2.438, 0.812, and 0.914. We therefore interpret the Stage-1 nuPlan gains as policy-learning improvements built on a strong representation start, not as an artifact of weak initialization.

To reduce the visual burden of these tables, Figure 4 is reserved for a compact summary of the Stage-0 stability pattern, the Stage-1 ranking reversal, and the dataset-statistics context that helps explain why the nuPlan regime favors WM-Decoupled-NoVis.

C. Downstream nuPlan Planner-Like Sanity Check

Table IV evaluates the two 50k nuPlan finalists with a downstream offline planner-like sanity probe. This is not an external closed-loop benchmark: it reuses the nuPlan validation split and checks whether the learned policy stays closer to teacher-derived planner actions while preserving the latent-rollout advantages observed in Stage 1. The value of this table is therefore diagnostic rather than definitive.

The planner-like probe supports the same main condition as the 20k/50k Stage-1 tables. WM-Decoupled-NoVis is not only better on latent return and collision, but also closer to the teacher-derived action target, reducing action MSE from 8.863 to 6.628. WM-Object shows a larger progress proxy, but this comes with worse action agreement and much higher collision rate, suggesting a more aggressive yet less planner-like policy. We therefore treat the planner-like probe as downstream support for the nuPlan ranking rather than as a separate benchmark claim.

Table V provides the most direct answer to *when* relation-aware abstraction helps. The advantage of WM-Decoupled-NoVis is not a global average artifact: it becomes larger exactly on the subsets where relation structure is most actionable. The clearest case is `lane_conflict`, where action MSE drops from 7.023 to 4.225, return rises from 0.943 to 13.330,

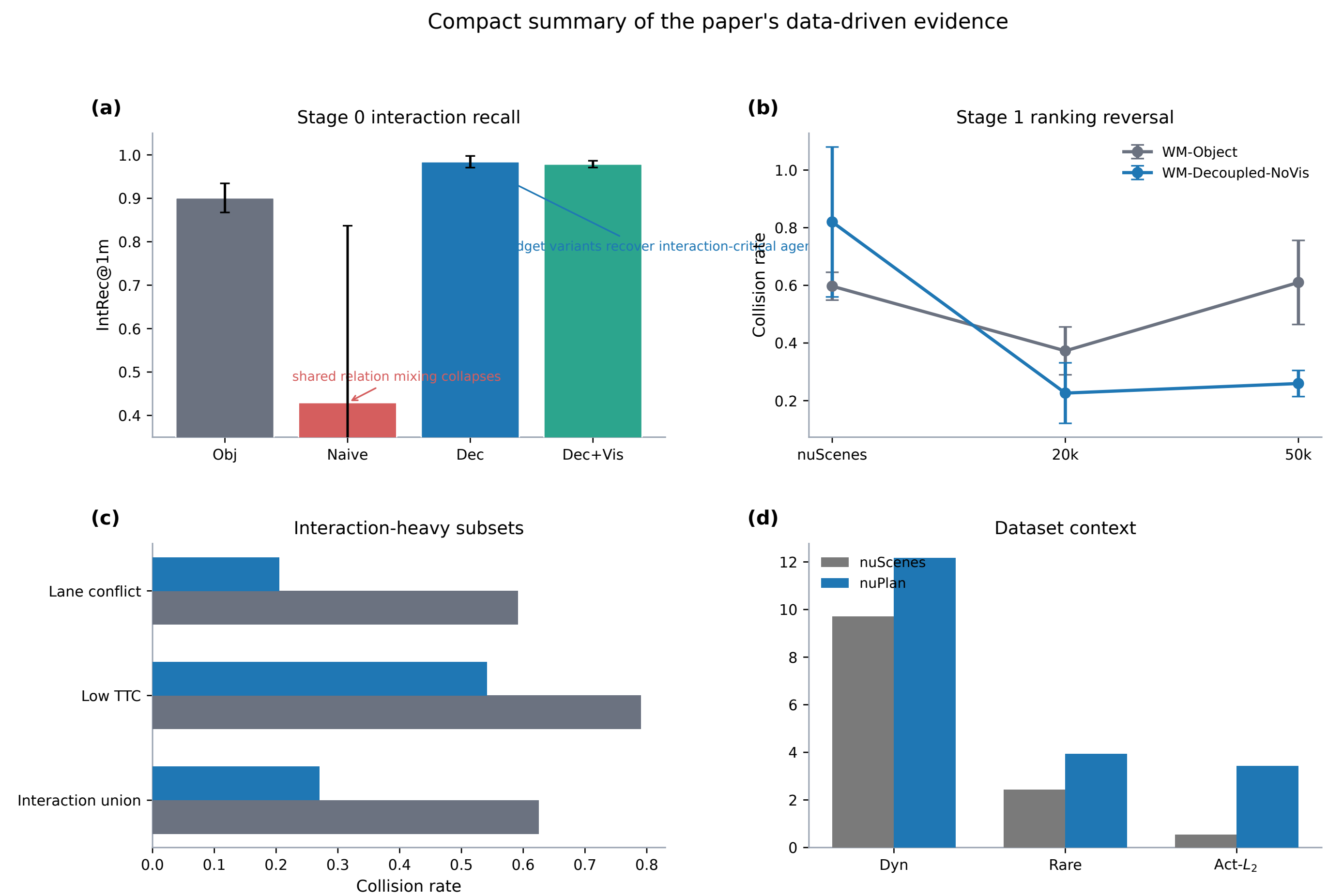

Fig. 4. Compact summary of the paper's data-driven evidence, drawn directly from the reported Stage-0, Stage-1, subset, and dataset-statistics aggregates. Panel (a) highlights the Stage-0 collapse of naive shared relation mixing and the recovery from typed-budget decoupling. Panel (b) shows the Stage-1 ranking reversal across nuScenes, nuPlan 20k, and nuPlan 50k. Panel (c) focuses on the interaction-heavy nuPlan subsets where WM-Decoupled-NoVis yields the largest collision reduction. Panel (d) summarizes the dataset-context differences that help explain why nuPlan makes relation structure more actionable.

TABLE IV
NUPLAN 50K OFFLINE PLANNER-LIKE SANITY CHECK. ALL RESULTS ARE AVERAGED OVER THREE SEEDS. LOWER ACTION ERROR INDICATES CLOSER AGREEMENT WITH THE TEACHER-DERIVED PLANNER ACTION.

Condition	Teacher Action MSE ↓	Action ΔL_2 ↓	Return ↑	Coll. Rate ↓	Progress Proxy ↑
WM-Object	8.863 ± 0.370	3.553 ± 0.118	1.722 ± 17.888	0.610 ± 0.146	2.995
WM-Decoupled-NoVis	$\mathbf{6.628 \pm 0.110}$	$\mathbf{2.115 \pm 0.083}$	$\mathbf{14.512 \pm 2.925}$	$\mathbf{0.259 \pm 0.045}$	1.082

TABLE V
NUPLAN 50K INTERACTION-CONDITIONED SUBSET ANALYSIS. EACH SUBSET ISOLATES SCENES WHERE RELATION REASONING SHOULD MATTER MOST. ALL VALUES ARE AVERAGED OVER THREE SEEDS.

Subset	Condition	Teacher Action MSE ↓	Return ↑	Coll. Rate ↓
lane_conflict	WM-Object	7.023	0.943	0.591
lane_conflict	**WM-Decoupled-NoVis**	**4.225**	**13.330**	**0.205**
low_ttc_proxy	WM-Object	12.796	2.650	0.791
low_ttc_proxy	**WM-Decoupled-NoVis**	**11.534**	**15.946**	**0.541**
high_interaction_union	WM-Object	8.836	1.588	0.625
high_interaction_union	**WM-Decoupled-NoVis**	**6.555**	**14.418**	**0.270**

and collision rate falls from 0.591 to 0.205. The same direction holds on `low_ttc_proxy`, `rare_agent_dense`, `dense_agents`, and their union, with collision reductions such as $0.791 \rightarrow 0.541$ for low-TTC scenes and $0.625 \rightarrow 0.270$ for the interaction-heavy union. We view this subset analysis as more explanatory than the frozen official closed-loop sanity table: the latter is still dominated by wrapper feasibility, whereas the subset probe shows directly that relation-aware abstraction matters most when lane conflict, low TTC, and dense interactive context are present.

Table VI further sharpens this interpretation. In pure cross-dataset rollout evaluation, a nuScenes-trained WM-Decoupled(+Vis) policy transfers poorly to nuPlan, dropping to -1.43 mean return with ego-step drift 0.853, whereas a nuPlan-trained WM-Decoupled-NoVis policy transfers to nuScenes better than a nuPlan-trained WM-Object policy, improving return from 5.51 to 16.07 and reducing collision rate from 0.624 to 0.434. In a separate nuScenes horizon-sensitivity study, the decoupled collision gap grows from $+0.113$ at rollout horizon $K = 3$ to $+0.306$ at $K = 7$. These follow-ups do not define new main benchmarks, but they make the paper's interpretation more complete: the nuScenes Stage-1 gap is tied to compounding imagination drift, whereas the nuPlan Stage-1 gain survives transfer-style probes and interaction-focused slices.

Figure 5 complements the tables with a more compact view of the downstream nuPlan evidence. Rather than introducing a new benchmark claim, it visualizes two already established facts: WM-Decoupled-NoVis is closer to the teacher-derived action target in the planner-like sanity probe, and its collision advantage becomes larger on the interaction-conditioned subsets where relation reasoning should matter most.

VII. Discussion

The main scientific lesson of the paper is that representation quality and policy-learning utility should be separated analytically. Stage 0 shows that typed-budget abstraction reliably improves decision-sufficient structure under a fixed latent budget. Stage 1 shows that this does not imply universal downstream gains. On nuScenes short-horizon imagination, the object-only policy remains the strongest baseline. On nuPlan 20k/50k, the robust winner is instead the no-visibility decoupled variant, and both the downstream planner-like sanity probe and the interaction-conditioned subset analysis support the same choice. The paper therefore argues against two simplistic narratives at once: neither "relation tokens are always helpful" nor "relation-aware abstractions do not work" is correct.

This cross-dataset ranking reversal suggests that the usefulness of relation-aware abstraction depends on the downstream planning regime and tokenization characteristics. In nuScenes Stage 1, relation information may be harder to exploit because the short-horizon latent objective, token noise, and actor landscape favor a simpler dynamic-only policy. In nuPlan 20k/50k, the preprocessed planning-oriented tokens appear to make relation structure more actionable, allowing the decoupled abstraction to translate representation gains into policy gains. The interaction-conditioned subset analysis sharpens this claim further: the gain is largest when lane conflict, low TTC, and dense interaction are present, exactly where relation-aware abstraction should matter most. Supporting dataset statistics are consistent with this view: nuPlan has higher dynamic-token density (12.164 versus 9.715 per sample), more rare agents (3.934 versus 2.439), nearly constant visibility (1.000 versus 0.746), and a much larger teacher-action scale (3.420 versus 0.539 in action L_2). These differences are not performance metrics, but they help explain why typed dyn/rel budgeting and the no-visibility condition become more useful on nuPlan than on nuScenes.

Two additional follow-up evaluations reinforce this interpretation. First, pure cross-dataset rollout evaluation shows that the ranking reversal is not just an artifact of evaluating each checkpoint on its native dataset. A nuScenes-trained WM-Decoupled(+Vis) policy transfers poorly to nuPlan, with mean return dropping to -1.43 versus 27.18 for nuScenes-trained WM-Object and ego-step drift rising to 0.853 versus 0.369. In the opposite direction, a nuPlan-trained WM-Decoupled-NoVis policy transfers to nuScenes better than a nuPlan-trained WM-Object policy, improving mean return from 5.51 to 16.07 and reducing collision rate from 0.624 to 0.434. Second, a nuScenes rollout-horizon sensitivity study shows that the decoupled collision gap grows from $+0.113$ at horizon $K = 3$ to $+0.306$ at $K = 7$, while ego-step drift remains high throughout. Together, these support results suggest that the nuScenes Stage-1 weakness is a compounding imagination-stability problem rather than a contradiction of the Stage-0 representation findings. This difference is itself a useful research result because it identifies the representation-to-policy interface as a dataset-dependent phenomenon rather than a universal monotonic ladder.

Several limitations remain. First, the paper still assumes oracle-token evaluation, so the results do not yet include perception noise. Second, Stage 1 studies imagination-based policy learning rather than a final external closed-loop benchmark, so the policy claims remain one step short of full benchmark evaluation. We have already connected official nuPlan closed-loop simulation through an oracle-token planner wrapper with corridor projection and local safety candidate scoring, but wrapper quality still dominates the outcome and we therefore do not treat those runs as main evidence. Third, the exact role of visibility remains unresolved: it behaves as a weak Stage-0 enhancement, a nuScenes Stage-1 stabilizer, and a nuPlan Stage-1 liability. Finally, although the cross-dataset and horizon-sensitivity probes strengthen our interpretation, they do not yet replace broader external baselines or a mature official closed-loop benchmark.

VIII. Conclusion

This paper presents DOOR, a decision-oriented object-relational typed-budget abstraction for imagination-based driving policy learning. The core technical point is that relation-aware latent models must allocate capacity by semantic role rather than naively mixing all tokens into one shared bottleneck. Under a fair 16-slot budget on nuScenes, the proposed abstraction consistently improves representation sufficiency

TABLE VI

SUPPORT DIAGNOSTICS BEYOND THE MAIN STAGE-1 TABLES. CROSS-DATASET TRANSFER ROWS ARE AVERAGED OVER THREE SEEDS. HORIZON-SENSITIVITY ROWS COME FROM THE SYNCHRONIZED NUSCENES FOLLOW-UP SWEEP AND ARE REPORTED AS DIRECT DIAGNOSTIC VALUES RATHER THAN A NEW BENCHMARK. LOWER EGO-COS STEP INDICATES SMALLER STEP-TO-STEP EGO-LATENT DRIFT.

Setting	Condition	Return ↑	Coll. Rate ↓	Ego-cos step ↓	Max action norm
nuScenes → nuPlan	WM-Object	27.18 ± 9.34	0.593 ± 0.213	0.369 ± 0.166	–
nuScenes → nuPlan	WM-Decoupled(+Vis)	-1.43 ± 14.11	0.593 ± 0.262	0.853 ± 0.083	–
nuPlan → nuScenes	WM-Object	5.51 ± 9.51	0.624 ± 0.110	0.301 ± 0.126	–
nuPlan → nuScenes	**WM-Decoupled-NoVis**	$\mathbf{16.07 \pm 1.79}$	$\mathbf{0.434 \pm 0.037}$	$\mathbf{0.181 \pm 0.032}$	–
nuScenes $K = 3$	WM-Object	11.856	0.570	0.296	3.781
nuScenes $K = 3$	WM-Decoupled	1.646	0.683	0.710	4.056
nuScenes $K = 5$	WM-Object	21.814	0.603	0.190	3.846
nuScenes $K = 5$	WM-Decoupled	-1.462	0.838	0.725	4.205
nuScenes $K = 7$	WM-Object	31.950	0.610	0.143	3.850
nuScenes $K = 7$	WM-Decoupled	-5.914	0.916	0.733	4.209

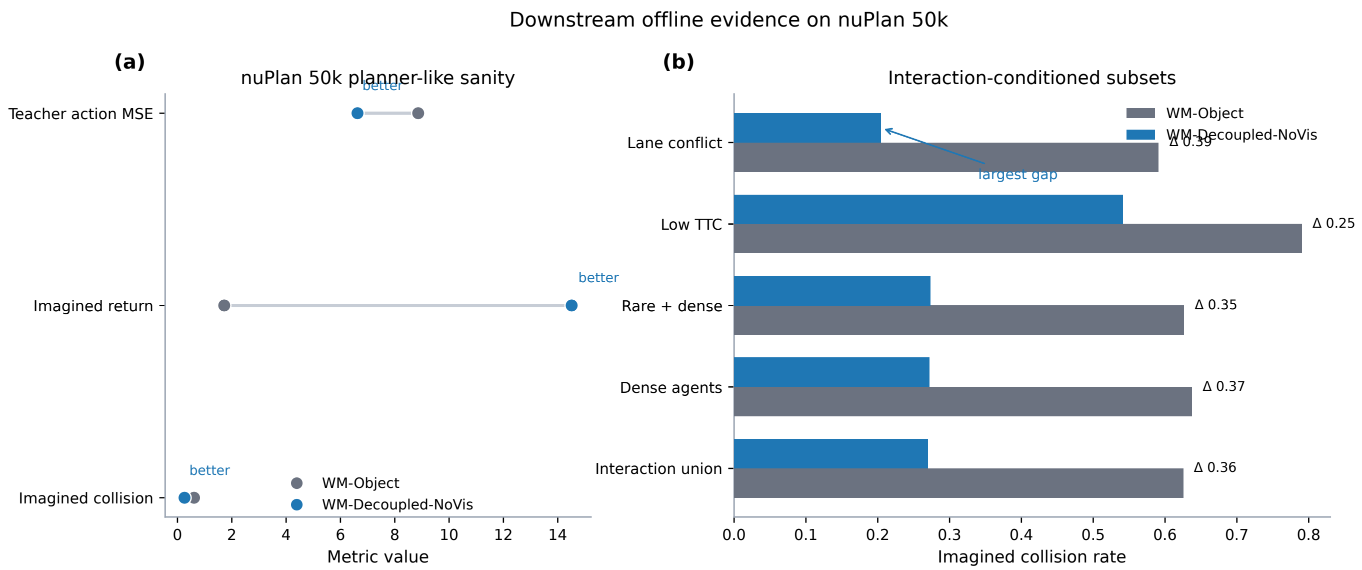

Fig. 5. Downstream offline evidence on nuPlan 50k, drawn directly from the planner-like sanity probe and the interaction-conditioned subset analysis. Panel (a) compares the two 50k finalists on teacher-action MSE, imagined return, and imagined collision. Panel (b) shows that the collision advantage of WM-Decoupled-NoVis is largest on subsets such as lane conflict, low TTC, dense agents, and the interaction union, which are exactly the regimes where relation-aware abstraction should matter most.

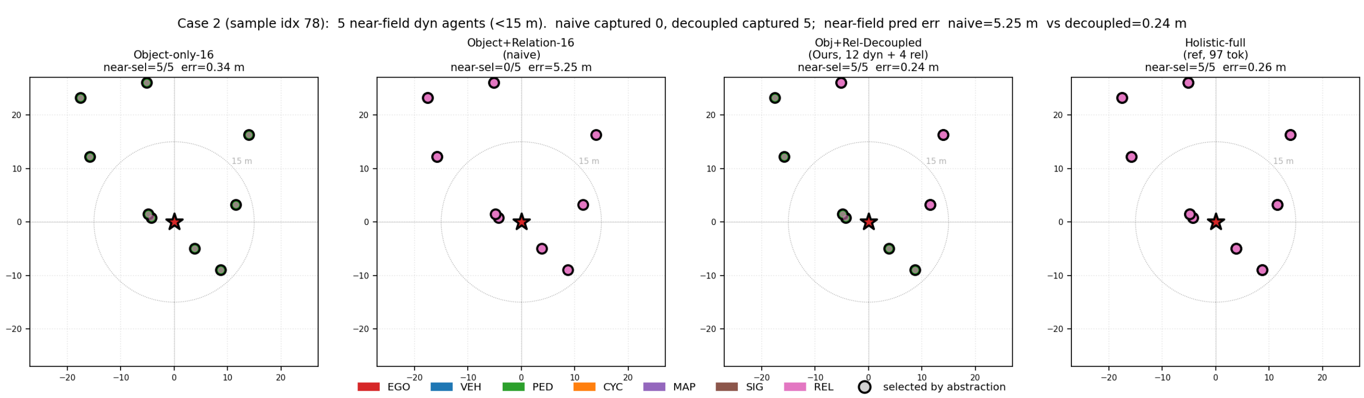

Fig. 6. Qualitative slot-selection case under the fair 16-slot budget. In this sample, the naive Object+Relation-16 variant captures none of the five near-field dynamic agents within 15 m and incurs 5.25 m near-field prediction error. The proposed Obj+Rel-Decoupled variant captures all five and reduces the error to 0.24 m, closely matching the high-context holistic reference while preserving the same compact budget as the naive model.

and resolves the catastrophic failure of naive relation mixing. However, the downstream policy-learning story is more nuanced: on nuScenes short-horizon imagination, object-only remains the most stable policy baseline, whereas on nuPlan 20k and its 50k scale-up the no-visibility decoupled variant becomes the robust winner and is further supported by offline planner-like probes, interaction-conditioned subsets, and cross-dataset transfer evidence. We therefore conclude that typed-budget relation-aware abstraction is a reliable representation improvement, but its downstream benefit depends on the planning regime and tokenization quality. We view that conditional result as the paper's main contribution: it turns a simple "does it work?" question into a sharper research statement about when relation structure becomes actionable for driving policy learning.

Appendix A
Planned Supplementary Material

The final submission should place four supplementary blocks here or in the companion supplemental document.

1) **Per-seed policy-learning tables:** full seed-level results for nuScenes Stage 1, nuPlan 20k, and nuPlan 50k.
2) **Dataset-statistics appendix:** full token-count, rare-agent, visibility, and teacher-action statistics used to explain the nuScenes-versus-nuPlan ranking reversal.
3) **Cross-dataset and horizon-sensitivity appendix:** pure transfer rollouts between nuScenes and nuPlan, together with the nuScenes $K = 3/5/7$ horizon-sensitivity study that diagnoses compounding imagination drift.
4) **Closed-loop sanity appendix:** official nuPlan wrapper variants, corridor-projection sanity results, the interaction-conditioned subset protocol, and a short explanation of why the external-loop runs are not treated as the main benchmark evidence yet.

References

[1] D. Ha and J. Schmidhuber, "World Models," *arXiv preprint arXiv:1803.10122*, 2018.

[2] D. Hafner, J. Pasukonis, J. Ba, and T. Lillicrap, "Mastering Diverse Domains through World Models," *arXiv preprint arXiv:2301.04104*, 2023.

[3] A. Dosovitskiy, G. Ros, F. Codevilla, A. Lopez, and V. Koltun, "CARLA: An Open Urban Driving Simulator," in *Proc. Conf. Robot Learning*, 2017.

[4] H. Caesar *et al.*, "nuScenes: A Multimodal Dataset for Autonomous Driving," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2020.

[5] H. Caesar *et al.*, "nuPlan: A Closed-Loop ML-Based Planning Benchmark for Autonomous Vehicles," *arXiv preprint arXiv:2106.11810*, 2021.

[6] J. Mao *et al.*, "Waymax: An Accelerated, Data-Driven Simulator for Large-Scale Autonomous Driving Research," *arXiv preprint arXiv:2310.08710*, 2023.

[7] D. Dauner *et al.*, "NAVSIM: Data-Driven Non-Reactive Autonomous Vehicle Simulation and Benchmarking," *arXiv preprint arXiv:2406.15349*, 2024.

[8] F. Locatello *et al.*, "Object-Centric Learning with Slot Attention," in *Adv. Neural Inf. Process. Syst.*, 2020.

[9] D. Chen, V. Koltun, and P. Krähenbühl, "Learning to Drive from a World on Rails," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, 2021.

[10] B. Kerbl, G. Kopanas, T. Leimkühler, and G. Drettakis, "3D Gaussian Splatting for Real-Time Radiance Field Rendering," *ACM Trans. Graph.*, vol. 42, no. 4, 2023.

[11] TODO, "Representative Reconstruction-Based Driving World-Model Baseline," replace this placeholder with the exact citation of the RAD-style comparison paper used in the final manuscript.

[12] TODO, "Representative 3DGS-Based Closed-Loop Driving Policy-Learning Paper," replace this placeholder with the exact citation of the reconstruction-based comparison paper used in the final manuscript.

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Second Author Second Author is a placeholder biography. Replace this paragraph with a short biography of the second author before the final T-IV submission.

Corresponding Author Corresponding Author is a placeholder biography. Replace this paragraph with a short biography of the corresponding author before the final T-IV submission.